PHASE 2: PROJECT DESIGN PHASE

Project Details

Particular	**Details**
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Phase	Project Design Phase
Date	31 January 2025
Team ID	________________
Prepared By	Koppana Navya

2.1 Data Flow Diagram (DFD) & User Stories

Introduction

The Project Design Phase defines how data flows through the system and how users interact with it. The solution is designed to transform raw food ordering data into meaningful visual insights using Tableau dashboards and stories. A Flask web application is used to provide browser-based access to the published dashboards.

Data Flow Diagram (DFD)

Context Level (Level-0 DFD)

```
+-------------------------+
|     Kaggle Dataset     |
| (CSV - Food Orders)    |
+-----------+-------------+
            |
            v
+-------------------------+
|    Tableau Desktop     |
| Data Cleaning & Analysis|
```

```
+-----------+-------------+
          |
          v
+-------------------------+
| Dashboard & Story       |
| KPI | Charts | Filters  |
+-----------+-------------+
          |
          v
+-------------------------+
| Tableau Public          |
| Publish Dashboard       |
+-----------+-------------+
          |
          v
+-------------------------+
| Flask Web Application   |
+-----------+-------------+
          |
          v
+-------------------------+
| End Users               |
| Restaurant Owners       |
| Analysts | Managers     |
+-------------------------+
```

Level-1 Data Flow

1. Download the Food Ordering Behaviour dataset from a trusted repository.

2. Load the CSV dataset into Tableau Desktop.
3. Validate records, column names, and data types.
4. Perform data cleaning and create calculated fields (Age Groups).
5. Develop visualizations and dashboards.
6. Create Tableau Story for business insights.
7. Publish the workbook to Tableau Public.
8. Embed the published dashboard into a Flask web application.
9. Users access dashboards through a web browser.

User Stories

User Type	Functional Requirement	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Business Analyst	Dataset Analysis	US-01	As a business analyst, I want to load the dataset into Tableau for analysis.	Dataset loads successfully without errors.	High	Sprint-1
Business Analyst	Data Cleaning	US-02	As a business analyst, I want to clean and prepare the dataset before visualization.	Missing values are handled and data is validated.	High	Sprint-1
Dashboard User	Dashboard Access	US-03	As a user, I want to view an interactive dashboard.	Dashboard loads with all charts.	High	Sprint-2
Dashboard User	Interactive Filters	US-04	As a user, I want to filter visualizations by city.	Charts update dynamically based on filter selection.	High	Sprint-2

User Type	Functional Requirement	User Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Restaurant Manager	Business Insights	US-05	As a restaurant manager, I want to identify popular cuisines and meal types.	Dashboard displays cuisine demand and meal trends.	High	Sprint-2
Delivery Manager	Delivery Analysis	US-06	As a delivery manager, I want to analyze delivery times.	Delivery performance chart is available.	Medium	Sprint-2
Administrator	Tableau Publishing	US-07	As an administrator, I want to publish dashboards to Tableau Public.	Dashboard is successfully published.	High	Sprint-3
Web User	Dashboard Integration	US-08	As a web user, I want to access the Tableau dashboard through a Flask application.	Dashboard is embedded and accessible via browser.	High	Sprint-3

2.2 Solution Requirements

Functional Requirements

FR No.	Functional Requirement	Sub Requirement (Story / Task)
FR-1	Dataset Collection	Download Food Ordering Behaviour dataset from Kaggle
FR-2	Dataset Loading	Import CSV dataset into Tableau Desktop
FR-3	Data Preparation	Clean missing values, validate data, create calculated fields

FR No.	Functional Requirement	Sub Requirement (Story / Task)
FR-4	Data Visualization	Develop KPI cards, charts, treemaps, donut charts, heat maps, and bar charts
FR-5	Dashboard Development	Design an interactive dashboard with filters and KPIs
FR-6	Story Development	Create Tableau Story containing multiple analytical scenes
FR-7	Tableau Public Publishing	Publish dashboards and stories online
FR-8	Flask Integration	Embed Tableau dashboard into a Flask web application

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Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	Dashboard should be simple, interactive, and easy to understand.
NFR-2	Security	Only trusted datasets should be used, and Tableau Public links should be shared securely.
NFR-3	Reliability	All dashboard visualizations should display accurate information without failure.
NFR-4	Performance	Dashboard should render all 50,000 records efficiently with minimal delay.
NFR-5	Availability	Dashboard should be available online through Tableau Public and Flask integration.
NFR-6	Scalability	The solution should support larger datasets and additional visualizations in future versions.

2.3 Technology Stack & System Architecture

Technical Architecture

```
Kaggle Dataset (CSV)
        │
        ▼
Tableau Desktop
(Data Cleaning & Visualization)
        │
        ▼
Tableau Dashboard & Story
        │
        ▼
Tableau Public Server
        │
        ▼
Flask Web Application
        │
        ▼
End Users
```

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web page for dashboard access	HTML5, CSS3
2	Application Logic	Web application routing	Python, Flask
3	Data Visualization	Interactive dashboards and stories	Tableau Desktop
4	Data Source	Food Ordering Behaviour dataset	CSV
5	Data Storage	Local dataset storage	Local File System

S.No	Component	Description	Technology
6	Dashboard Hosting	Publish dashboards online	Tableau Public
7	File Storage	Dataset and project files	Local Storage
8	External Platform	Dataset repository	Kaggle
9	Browser	Dashboard visualization	Google Chrome / Microsoft Edge
10	Analytics Tool	Data analysis and calculated fields	Tableau
11	Deployment	Local web server	Flask Development Server

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Framework	Lightweight web framework	Flask
2	Security	Controlled local file access and secure Tableau Public sharing	Flask, Tableau Public
3	Scalable Architecture	Modular dashboard design allows additional datasets and visualizations	Tableau + Flask
4	Availability	Dashboards accessible through Tableau Public and browser	Tableau Public
5	Performance	Efficient rendering of 50,000 records using Tableau optimization	Tableau Desktop

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2.4 Design Summary

Design Objectives

The project design aims to provide an end-to-end analytical solution that converts raw food ordering data into meaningful visual insights. The architecture combines Tableau for data visualization with Flask for lightweight web deployment.

The design emphasizes:

- Structured data processing.
- Interactive dashboard development.
- Business storytelling.
- User-friendly navigation.
- Online accessibility through Tableau Public.
- Future scalability for additional datasets and analytical features.

Expected Deliverables

- Data Flow Diagram (Context and Level-1)
- User Stories
- Functional Requirement Specification
- Non-Functional Requirement Specification
- Technical Architecture
- Components & Technologies Table
- Application Characteristics Table
- Architecture Summary

Conclusion

The Project Design Phase establishes the technical foundation of the Food Ordering Behaviour and Consumer Trends project. It defines the flow of data from dataset acquisition to visualization and web deployment, specifies system requirements, identifies user interactions, and documents the technologies required for implementation. This structured design ensures that the project is scalable, maintainable, and capable of delivering interactive business intelligence solutions for restaurant owners, food delivery platforms, and decision-makers.